Characterizing entanglement in 1D many body systems

Namit Anand[1]

[1]National Institute of Science Education and Research, Bhubaneswar, India

E-mail: namit.anand@niser.ac.in

Abstract. The aim of the project is to characterize entanglement in 1D many body systems(fermions and bosons).

1. Introduction

In this project we have tried to characterize entanglement in 1D many body systems enabling it to capture the essential physics of 1 dimensional systems as well as enhancing theoretical calculations which can be directly tested in experiments.
The system of interest was ultracold bosons trapped in optical lattices since such systems display extremely precise control over the Hamiltonian and initial state preparation [1]. Hence such systems are excellent canditdates for a many body quantum computer.

Entanglement is a fundamental resource for quantum information processing and is present in the ground state of several naturally occurring many-body systems at low temperatures. Its not just the presence of entanglement that is important but also its scaling with the size of system partitions that determines the complexity of the quantum many-body system. A experimental estimation of entanglement in many-body systems is a tedious task, as it requires either full-state tomography(which scales exponentially) or the assumption of the Hamiltonian of the system or temperature etc.

However we suffer two big disadvantages namely all the parameters in the Hamiltonian are not known exactly and also the state rho is not known(except in some exactly solvable models) else entanglement estimation would be very easy. So the aim is to use experimentally measurable observables at our hand like quasi momentum, number density of particles, ground state energy etc. to estimate lower bounds to the total entanglement of the system.

We also take a few fermionic and bosonic models and calculate the entanglement entropy of such systems with a closed form expresiion and draw a phase diagram for entanglement entropy as a function of the interaction parameter.

2. Explorations

Areas explored in the project:

- Since one of the main parts of the project is entanglement characterization of bosons in optical lattices, I started with an excellent paper by M. Plenio on spatial entanglement of bosons in optical lattices [5]
  The paper uses a simple but powerful principle in combination with methods from optimization theory: "given a set of observables, we consider all density matrices that are compatible with experimentally measured expectation values of these observables. Among these density matrices, we find the one with the least amount of entanglement as quantified by a suitable entanglement measure." In this way, they determine a lower bound on the entanglement that must have been present in the state that gave rise to the observed expectation values. These bounds do not require any other assumptions on the system.
- I have encountered a very interesting aspect of entanglement: how to identify entanglement in indistinguishable particles. [6]
- I am reading the following paper by Vlatko Vedral: doi:10.1098/rspa.2007.1887, which talks about occupation number entanglement, which we need in order to look at entanglement in fermionic systems and since for bosons its argued that observables must commute with local number operator N. The paper suggests a situation in which one can overcome the selection rules for occupation number !
- Monogamy inequality:
  The amount of quantum correlations that can be shared amongst the subsystems of a multipartite quantum state is constrained by the idea of monogamy. For the tripartite case:
  if $\mathcal{C}$ is a bipartite quantum correlation measure, then this measure is said to be monogamous (or satisfy monogamy) for a tripartite quantum state ρ_{ABC}, and with A as the "nodal observer", if

$$\mathcal{C}(\rho_{\mathcal{A}:\mathcal{BC}}) \geq \mathcal{C}(\rho_{\mathcal{AB}}) + \mathcal{C}(\rho_{\mathcal{AC}}). \tag{1}$$

  Here $\mathcal{C}(\rho_{\mathcal{AB}})$ is the quantum correlation (with respect to the measure $\mathcal{C}$) between subsystems A and B, $\mathcal{C}(\rho_{\mathcal{AC}})$ is the quantum correlation between subsystems A and C, and $\mathcal{C}(\rho_{\mathcal{A}:\mathcal{BC}})$ is quantum correlation between subsystem A and subsystems B and C taken together. By rearranging the terms in the above inequality, we get

$$\delta\mathcal{C} \equiv \mathcal{C}(\rho_{\mathcal{A}:\mathcal{BC}}) - \mathcal{C}(\rho_{\mathcal{AB}}) - \mathcal{C}(\rho_{\mathcal{AC}}) \geq \prime. \tag{2}$$

  This leads to the concept of quantum monogamy score, which, for a given bipartite quantum correlation measure, is defined as

$$\delta\mathcal{C} \equiv \mathcal{C}(\rho_{\mathcal{A}:\mathcal{BC}}) - \mathcal{C}(\rho_{\mathcal{AB}}) - \mathcal{C}(\rho_{\mathcal{AC}}), \tag{3}$$

3. Superselection rules:

3.1. Flaw in the proof

4. Proof

In Ref. EJP paper, section 2.4(Impossibility of superposing states belonging to different coherent sectors), the authors construct a proof using a superoperator G that commutes with all observables of the system. The eigenstates of operator G are of the form $|g_m; \alpha_m\rangle$ where g_m and α_m are labels to distinguish different eigenvectors all of which are non-degenerate, i.e. $\langle g_m; \alpha_m | g_n; \alpha_n\rangle = 0$ if $m \neq n$.
In the next section(2.4), they construct a vector $|u\rangle = \sum_m u_m |g_m; \alpha_m\rangle$ superposing the eigenstates of G and claim that since G commutes with every other observable, $|u\rangle$ should be an eigenstate of every operator in a complete set of commuting observables of the system.
Now we know that **linear combination of non-degenrate eigenvectors is not an eigenvector** i.e. take a superposition of two eigenvectors of G, $|u\rangle = a|g_m; \alpha_m\rangle + b|g_n; \alpha_n\rangle$ where $a \neq b$ in general, then:

$$G|u\rangle = G(a|g_m; \alpha_m\rangle + b|g_n; \alpha_n\rangle) = a * g_m |g_m; \alpha_m\rangle + b * g_n |g_n; \alpha_n\rangle \tag{4}$$

Note that this equation isn't an eigenvalue equation for G, since $\mathrm{G}|u\rangle \neq \lambda|u\rangle$ unless $a = b$ which isn't true in general or $g_m = g_n$ which isn't true from the fact that the eigenstates of G are non-denegerate.

It is true that if G commutes with every other observable and has non-degenerate eigenvalues, then all such obervables share the same eigenvectors. But $|u\rangle$ itself is not an eigenvector of G as proved above. And so $|u\rangle$ cannot be an eigenvector of other observables too. To summarize, the proof of superselection rules proposed is thus is not valid for the case discussed by the authors. Whether a similar proof can be constructed is still open.

5. Plan of action

It turns out that density functional theory has linked entanglement and quantum phase transitions using HK theorem. http://dx.doi.org/10.1103/PhysRevA.74.052335 Another paper mentions how one can use local density approximations to calculate very accurately the entanglement in spatially inhomogeneous many fermion systems modelled using a Bose-Hubbard hamiltonian with inhomogeneous terms using just the knowledge of the n(r) and the Bethe-Ansatz integral equations. They have even shown an analytical form of the entanglement entropy calculations. This means that we can directly do these calculations and get the entanglement. Also can we estimate the concurrence which is experimentally measurable and cross check? http://journals.aps.org/prl/abstract/10.1103/PhysRevLett.100.070403

3. Also the characterization of entanglement in Fermionic systems has only been analyzed in 1D, can we do the same for higher dimensions? http://journals.aps.org/prl/abstract/10.1103/PhysRevLett.110.046806

1. We wish to use the analytic solution for the Lieb-Liniger model, Bethe ansatz and use the information of the ground state number density along with the entanglement entropy formula to estimate how it varies as a function of n,g.

I am using the analytical form for the energy density from this paper: http://iopscience.iop.org/1751-8121/44/10/102001/ I have done these, however as I plot the entanglement entropy as a function of n and g, it shows a strange behaviour which I will show you tomorrow. Kindly suggest other references from where I can try to use the form.

2. We would also like to use experimentally measurable observables like visibility to estimate entanglement and propose new bounds. I think we can do this analytically using the following paper : http://dx.doi.org/10.1103/PhysRevLett.95.220402 I will talk about this tomorrow.

3. The paper by Vedral is very interesting, I haven't read it in detail yet but if I can I will present tomorrow. http://journals.aps.org/pra/abstract/10.1103/PhysRevA.80.022338

4. The paper on momentum space entangled in QFTs needs to be explored for ideas. http://journals.aps.org/prd/abstract/10.1103/PhysRevD.86.045014

6. Connection between Entanglement entropy and Ground state energy density

7. Calculations on Entanglement Entropy for Yang Gaudin Model

So the Energy density as showin in Ref. [2] for the BCS like pairing and tightly bound molecules goes as:

$$E = \left(\frac{1}{2}gmn^2 + \frac{\pi^2 n^3}{12}\right) \tag{5}$$

Now the entanglement entropy for a fermionic gas goes as:
$\epsilon = -\frac{\partial E}{\partial g}\log_2\left(\frac{\partial E}{\partial g}\right) - 2\left(\frac{n}{2} - \frac{\partial E}{\partial g}\right)\log_2\left(\frac{n}{2} - \frac{\partial E}{\partial g}\right) - \left(\frac{\partial E}{\partial g} - n + 1\right)\log_2\left(\frac{\partial E}{\partial g} - n + 1\right)$
as quoted in Ref. [3]

For the above case this reduces to:

$$-\frac{mn^2 \log\left(\frac{mn^2}{2}\right)}{2\log(2)} \tag{6}$$

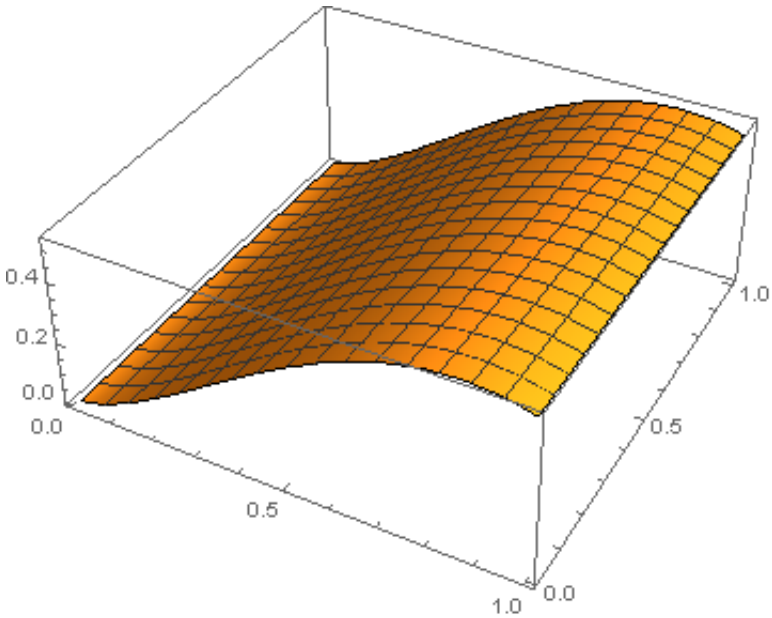

Figure 1. Entanglement entopy vs n and m

So the Energy density as showin in Ref. [2] for theYang Gaudin model in the strong repulsion limit for P = 0 is:

$$E = \frac{1}{3}\pi^2 n^3 \left(\frac{\pi^2 \zeta(3)}{\gamma^3} - \frac{32\log^3(2)}{\gamma^3} + \frac{12\log^2(2)}{\gamma^2} - \frac{4\log(2)}{\gamma} + 1 \right) \tag{7}$$

Now the entanglement entropy for a fermionic gas goes as:

$$\epsilon = \left(-2\left(\frac{n}{2} - \left(\frac{\alpha\frac{\partial E}{\partial \gamma}}{n} + \frac{\alpha\frac{\partial E}{\partial n}}{\gamma} \right) \right) \log_2 \left(\frac{n}{2} - \left(\frac{\alpha\frac{\partial E}{\partial \gamma}}{n} + \frac{\alpha\frac{\partial E}{\partial n}}{\gamma} \right) \right) - \left(\left(\frac{\alpha\frac{\partial E}{\partial \gamma}}{n} + \frac{\alpha\frac{\partial E}{\partial n}}{\gamma} \right) - n + 1 \right) \log_2 \left(\left(\right. \right. \right.$$
$$\left(- \left(\frac{\alpha\frac{\partial E}{\partial \gamma}}{n} + \frac{\alpha\frac{\partial E}{\partial n}}{\gamma} \right. \right. \tag{8}$$

as quoted in Ref. [3]

here $\alpha = \frac{1}{2}$

For the above case this reduces to:

$$\frac{\pi^2 n^2 \left(-3\gamma^2 + 8\gamma\log(2) - 12\log^2(2)\right) \log\left(\frac{\pi^2 n^2 \left(3\gamma^2 - 8\gamma\log(2) + 12\log^2(2)\right)}{6\gamma^3} \right)}{\gamma^3 \log(64)} \tag{9}$$

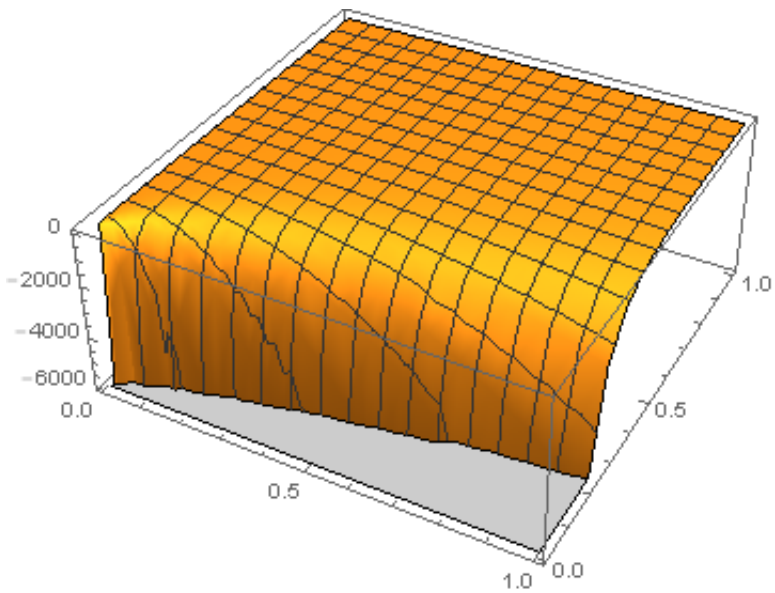

Figure 2. No data points seem to occur although there are no divergences

Energy density for the Yang Gaudin model in the strong repulsion limit for $P > 0.5$ goes as:

$$E = \frac{1}{3}\pi^2 n^3 \left(-\frac{8\left(32(n-2m)^3 - \frac{16}{5}\pi^2 n^2 (n-2m)\right)}{g^3} + \frac{48(n-2m)^2}{g^2} - \frac{8(n-2m)}{g} + 1 \right) \tag{10}$$

Now the entanglement entropy for a fermionic gas goes as:

$$\epsilon = -\left(-\frac{\partial E}{\partial g} - m + \frac{n}{2}\right)\log_2\left(-\frac{\partial E}{\partial g} - m + \frac{n}{2}\right) - \left(-\frac{\partial E}{\partial g} + m + \frac{n}{2}\right)\log_2\left(-\frac{\partial E}{\partial g} + m + \frac{n}{2}\right) - \left(\frac{\partial E}{\partial g} - n + 1\right)\log_2\left(\frac{\partial E}{\partial g} - n + 1\right) - \frac{\partial E}{\partial g}\log_2\left(\frac{\partial E}{\partial g}\right)$$

For the above case this reduces to: $-\frac{\log\left(\frac{1}{24}\pi^2\left(-\frac{24\left(2m+\frac{1}{2}(2m-n)+n\right)^2}{g^3} - \frac{4\mathrm{Abs}'(g)\left(2m+\frac{1}{2}(2m-n)+n\right)}{|g|^2} + \frac{32\left(-\frac{1}{4}11\pi^2(n-2m)^3\right.}{}\right.\right.}{}$

$$-\frac{\log\left(\frac{1}{24}\pi^2\left(-\frac{24\left(2m+\frac{1}{2}(2m-n)+n\right)^2}{g^3} - \frac{4\mathrm{Abs}'(g)\left(2m+\frac{1}{2}(2m-n)+n\right)}{|g|^2} + \frac{32\left(-\frac{1}{4}11\pi^2(n-2m)^3+\frac{15}{8}(n-2m)^3+\frac{63}{8}(2m+n)\pi^2(n-2m)^2-\frac{45}{4}(2m+n)(n-2m)^2+\right.}{5|g|}\right.\right.}{}$$

$$-\frac{\log\left(-\frac{1}{24}\pi^2\left(-\frac{24\left(2m+\frac{1}{2}(2m-n)+n\right)^2}{g^3} - \frac{4\mathrm{Abs}'(g)\left(2m+\frac{1}{2}(2m-n)+n\right)}{|g|^2} + \frac{32\left(-\frac{1}{4}11\pi^2(n-2m)^3+\frac{15}{8}(n-2m)^3+\frac{63}{8}(2m+n)\pi^2(n-2m)^2-\frac{45}{4}(2m+n)(n-2m)^2-\right.}{5|}\right.\right.}{}$$

$$-\frac{\log\left(-\frac{1}{24}\pi^2\left(-\frac{24\left(2m+\frac{1}{2}(2m-n)+n\right)^2}{g^3} - \frac{4\mathrm{Abs}'(g)\left(2m+\frac{1}{2}(2m-n)+n\right)}{|g|^2} + \frac{32\left(-\frac{1}{4}11\pi^2(n-2m)^3+\frac{15}{8}(n-2m)^3+\frac{63}{8}(2m+n)\pi^2(n-2m)^2-\frac{45}{4}(2m+n)(n-2m)^2-\right.}{5|}\right.\right.}{}$$

yang4.png

Figure 3. No graph due to divergences

Energy density for the Yang Gaudin model in the strong attraction limit goes as:

$$E = \frac{1}{24}\pi^2(n-2m)^3 \left(\frac{4\left(\frac{1}{2}(2m-n)+2m+n\right)}{|g|} - \frac{32\left(-\frac{1}{4}11\pi^2(n-2m)^3+\frac{15}{8}(n-2m)^3+\frac{63}{8}\pi^2(2m+n)\right.}{} \right. \tag{11}$$

Now the entanglement entropy for a fermionic gas goes as:

$$\epsilon = -\left(-\frac{\partial E}{\partial g} - m + \frac{n}{2}\right)\log_2\left(-\frac{\partial E}{\partial g} - m + \frac{n}{2}\right) - \left(-\frac{\partial E}{\partial g} + m + \frac{n}{2}\right)\log_2\left(-\frac{\partial E}{\partial g} + m + \frac{n}{2}\right) - \left(\frac{\partial E}{\partial g} - n + 1\right)\log_2\left(\frac{\partial E}{\partial g} - n + 1\right) - \frac{\partial E}{\partial g}\log_2\left(\frac{\partial E}{\partial g}\right)$$

For the above case this reduces to: $\frac{(2m-n)\left(15g^4-80\pi^2n^3\left(g^2+12g(2m-n)+96(n-2m)^2\right)+768\pi^4n^5\right)\log\left(\frac{(2m-n)\left(-15g^4+8\right.}{}\right.}{}$

yang3.png

Figure 4. Entanglement entopy vs n and m

8. Conformal Field theory for Dipolar gas

8.1. Introduction

Beside its own fundamental theoretical interest, a principal reason for the success of the entanglement entropy as an entanglement measure in extended quantum systems is surely its universal scaling at one-dimensional (1D) conformal critical points. The equation

$$S_A = \frac{c}{3} \ln \frac{\ell}{a} + c_1' \,, \tag{12}$$

has become one of the most ubiquitous formulas in the last five years' literature, appearing in fields as apparently unrelated as quantum information, condensed matter, and high energy physics. The reasons for this prominence are clear: it is a single quantity, easily measurable in numerical simulations, that at the same time gives the location of the critical point and one of its most important universal signatures, the *central charge* c of the underlying conformal field theory (CFT).

8.2. Basic definitions

Let ρ be the density matrix of a system, which we take to be in the pure quantum state $|\Psi\rangle$, so that $\rho = |\Psi\rangle\langle\Psi|$. Let the Hilbert space be written as a direct product H= $H_A \otimes H_B$. A's reduced density matrix is $\rho_A = \mathrm{Tr}_B\, \rho$. The entanglement entropy is the corresponding von Neumann entropy

$$S_A = -\,\mathrm{Tr}\ \rho_A \ln \rho_A \,, \tag{13}$$

and analogously for S_B. When ρ corresponds to a pure quantum state $S_A = S_B$. For future use, we also define the Rényi entropies

$$S_A^{(n)} = \frac{1}{1-n} \ln \mathrm{Tr}\, \rho_A^n \,, \tag{14}$$

that also are characterized by $S_A^{(n)} = S_B^{(n)}$ whenever ρ corresponds to a pure quantum state. From these definitions $S_A = \lim_{n\to 1} S_A^{(n)}$. Ref. [1]

9. Particle partition entanglement for a Dipolar gas

Entanglement entropy of many-body systems is most often studied under spatial bipartitions. For example, if a translationally invariant 1D system is bipartitioned into contiguous subsystems, then subsystem A is completely characterized by the length of the subsystem ℓ_A. For the ground states of critical 1D systems, the leading order scaling of the EE is logarithmic in ℓ_A,

$$S\left(\ell_A\right) \approx \frac{c}{3} \log \ell_A + \ldots \,, \tag{15}$$

where c is the central charge of the underlying conformal field theory. In systems of itinerant particles, one may alternatively bipartition the system into subsets of particles

(Fig.), rather than spatial subregions. An indistinguishable particle partition is uniquely characterized by the number of particles n in the subsystem, with the RDM for the subsystem given by the n-body reduced density matrix:

$$\rho_n \equiv \int d\boldsymbol{r}_n \ldots \int d\boldsymbol{r}_{N-1} \langle \boldsymbol{r}_n \ldots \boldsymbol{r}_{N-1} | \rho_N | \boldsymbol{r}_n \ldots \boldsymbol{r}_{N-1} \rangle \tag{16}$$

where $\{\boldsymbol{r}_i\}$ are the positions of the particles and ρ_N is the full N-body density matrix. One can then quantify the entanglement entropies between subsets of particles by the particle partitioned EE:

$$S_\alpha(n) \equiv S_\alpha[\rho_n]. \tag{17}$$

Note that since ρ_n is non-local in space and contains no length scale, $S_\alpha(n)$ can capture physics that is distinct from, and complimentary to, the EE under a spatial bipartition.

The importance of particle partition EE, and in particular, its connection to physically utilizable entanglement for quantum information processing applications has only recently been uncovered . Zozulya *et al.* have proposed a general scaling form for particle EE which is linear in the subsystem size and logarithmic in the system size :

$$S(n, N) = b_1 n \log N + b_2. \tag{18}$$

The coefficients b_1, b_2 have only been computed in a limited number of models and one may ask what physical content is contained within. Additionally, previous work has demonstrated that in systems of itinerant bosons, the $n = 1$ second ren EE is bounded by the experimentally accessible condensate fraction, thus offering a direct relationship between particle EE and experimentally measurable quantities.

10. One particle entanglement in bosonic Luttinger Liquids

10.1. Asymptotic scaling for $L \to \infty$

We begin with an analysis of the one particle partition 2nd ren EE $S_2(n = 1)$ for a LL described by . Using the Galilean invariance of the one-body RDM, we take the general, appropriately normalized ($\mathrm{Tr}\,\rho_1 = 1$) and regularized form of ρ_1 to be: where we have used , a is some short distance length scale and $\Phi(y)$ is a dimensionless function satisfying $\Phi(y) \leq 1$, $\Phi(0) = 1$ that depends on the non-universal details of the microscopic model. Performing the trace in $\alpha = 2$ yields where

$$\overline{\Phi^2}(y) \equiv y^{-1} \int_0^y \Phi(z)^2 dz. \tag{19}$$

Taking the log, the central finding of our work is that the finite size scaling of the one-particle 2nd ren EE for $K > 1$ is described by where $b_1 = 1/K$ and the constants $b_{2,3}$ depend on both K as well as the short distance physics captured in $\Phi(r/a)$. We note this result is consistent with previously reported results for the Tonks-Girardeau model of impenetrable bosons, which corresponds to $K \to 1$ where it was found that

$S_1(n = 1) \sim \ln N$. Thus the leading scaling coefficient of the particle EE is equal to the inverse of the Luttinger parameter which characterizes the universal quantum hydrodynamics in one dimension.

References

[1] Jaksch, D., Bruder, C., Cirac, J. I., Gardiner, C. W. & Zoller, P. Cold bosonic atoms in optical lattices. Phys. Rev. Lett. 81, 31083111 (1998).
[2] http://arxiv.org/abs/0905.4013v2
[3] Fermi gases in 1d
[4] http://journals.aps.org/pra/abstract/10.1103/PhysRevA.74.042325
[5] http://www.nature.com/ncomms/2013/130718/ncomms3161/abs/ncomms3161.html
[6] 10.1103/PhysRevLett.91.097902